

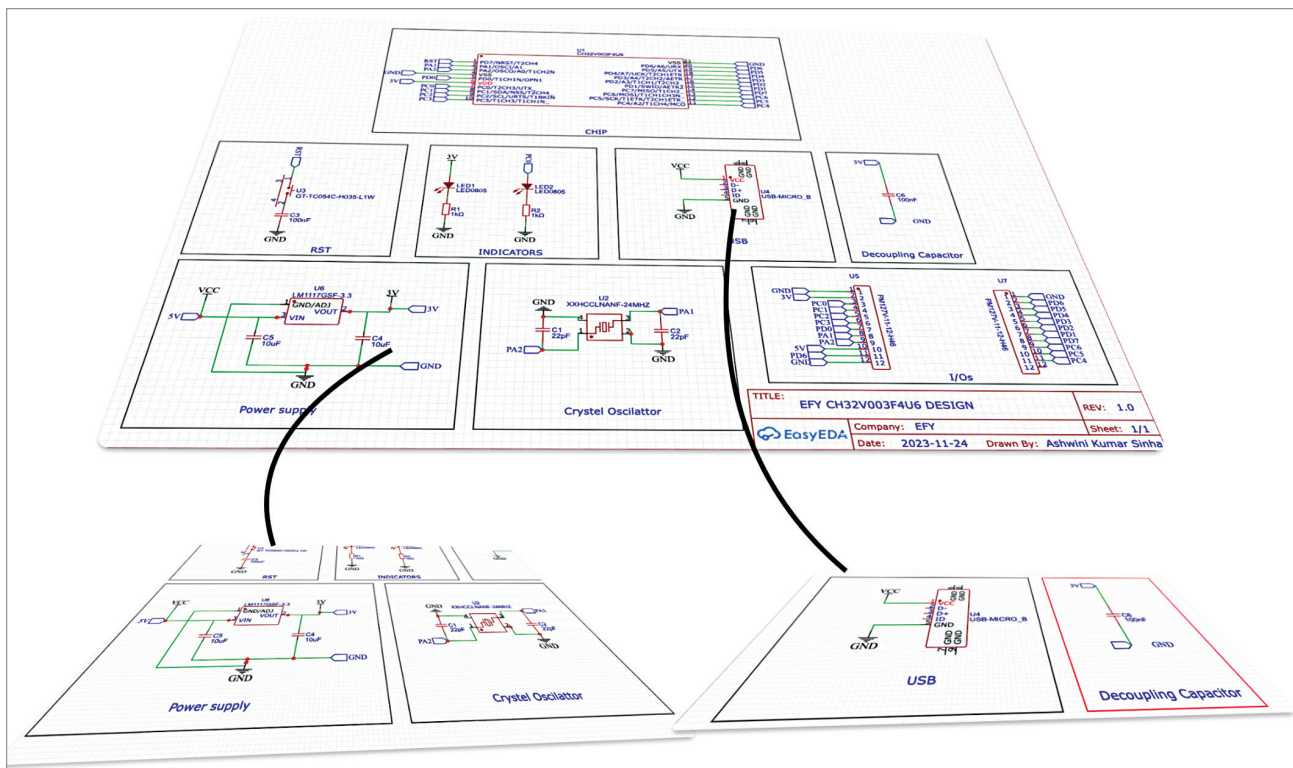


Fig. 4: Example circuit of RISC-V system where each component is divided into blocks connected over nets

connection lines attached to it get deleted, and you need to rewire it. In the second diagram, you can simply delete and use another component if needed to modify the design. Only the net port gets deleted, and you can easily add the new nets and connect the component using nets.

What are nets

Nets, in the context of electronics design, refer to logical connections between different components in a circuit. Instead of physically drawing wires between components, engineers use the named nets to represent these connections.

Why do professional design engineers use nets?

1. Clarity and organisation.

Nets provide a clear and organised representation of connections within a circuit. By naming and labelling nets appropriately, engineers can easily understand the interconnections between components. A big

and complex circuit with many components connected using wires can become messy and unreadable, with lines crossing each other. Using nets allows engineers to separate these connections.

2. Modularity and scalability.

Using nets facilitates a modular design approach, where circuits are broken down into manageable sections or modules. This modularity promotes scalability, allowing engineers to reuse modules in different designs or expand the circuit without redesigning the entire system (Fig. 3).

3. Ease of modification.

Nets-based designs are easier to modify and maintain compared to wiring-based designs. When changes are needed, engineers can simply modify the connections between nets without having to redraw the entire schematic. This flexibility streamlines the design iteration process and reduces the risk of errors.

4. Hierarchical design. Nets enable hierarchical design methodologies, where complex circuits are divided into hierarchical blocks or modules. Each module communicates with others via named nets, simplifying the overall design structure and enhancing readability (Fig. 4).

5. Compatibility with EDA tools. Electronic design automation (EDA) tools, used for schematic capture and PCB layout, support nets-based designs. These tools can automatically generate netlists, which describe the connections between components, facilitating the design-to-production workflow.

6. Standardisation and collaboration. Using nets promotes standardisation and consistency in circuit designs. Engineers can easily share schematics with colleagues or collaborators, knowing that the logical connections represented by nets will be universally understood.